Take home Final Exam PHYS 508

Ground Rules: Do the work all by yourself, and show it in full. You can use textbooks, class notes, Mathematica or analogous software. But don't use AI/LLM software like ChatGPT or Gemini etc. Be prepared to explain all of your work if it were necessary.

If you have questions or find errors in the problems just email me at **bunker@illinoistech.edu**.

Please email the completed exam to me by **noon on Sunday Dec 14, 2025**. If you need to hand it in on paper just put it in my departmental mailbox, but let me know by email that you did so.

have fun! gb

Problem 1 Inertia tensor

A uniform cone of mass M, radius R and height H is suspended from its vertex in gravitational acceleration g on a frictionless pivot as shown in the figure below. It is displaced from the vertical so it swings back and forth.

a) write down a Lagrangian for its motion. Do not make the small angle approximation.

b) What is its frequency of small oscillations? Express your result in terms of R, H, and g.

(auxiliary) In[∘]:=

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Graphics3D[{Thickness[.01], Line[{{0, -1/2, 1}, {0, 1/2, 1}}],
  Opacity[.3], Cone[{{0, 0, 0}, {0, 0, 1}}, 1]}]
```

(auxiliary) Out[∘]=

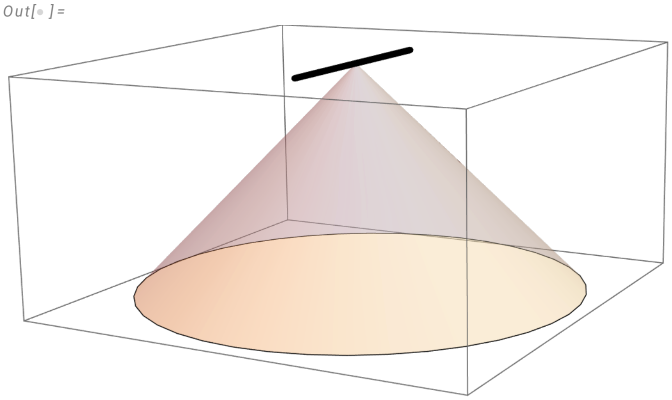

figure 1a

b) if it were not swinging, but were spun around its axis of symmetry at angular velocity ω, as indicated in figure 1b, what would be its angular momentum, and its kinetic energy?

(auxiliary) In[ ]:=

```
Graphics3D[{Thickness[.01], Arrow[{{0, 0, 0}, {0, 0, 1}}],
  Opacity[.3], Cone[{{0, 0, 1}, {0, 0, 0}}, 1]}]
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(auxiliary) Out[ ]=

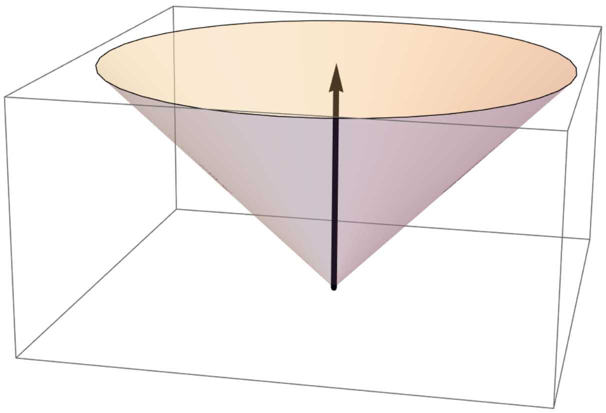

figure 1b

c) if the cone were spun around an axis that ran from the vertex along the inclined side (as shown in the figure 1c below), at angular velocity ω, what would be its *angular momentum*, and its *kinetic energy*?

(auxiliary) In[ ]:=

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Graphics3D[{Thickness[.01], Arrow[{{0, 0, 0}, {1, 0, 1}}],
  Opacity[.3], Cone[{{0, 0, 1}, {0, 0, 0}}, 1]}]
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(auxiliary) Out[ ]=

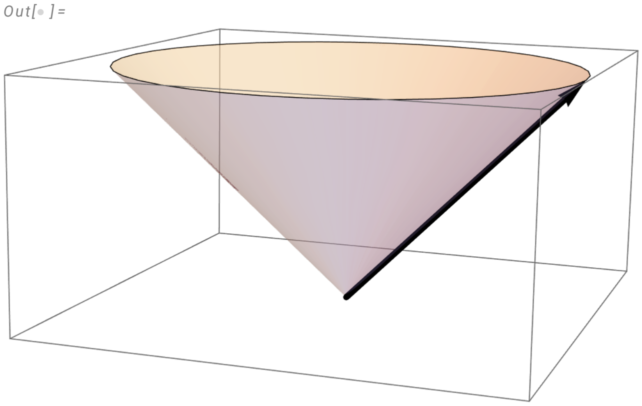

figure 1c

problem 2 Hamiltonian formulation

For the system described in problem 1a (swinging cone):

a) determine the generalized momentum that is conjugate to the angle θ of deviation of cone axis from vertical.

b) Write down the Hamiltonian, expressed in the terms of the angle and generalized momentum.

c) write down Hamilton's equations of motion for this system

d) make the small angle approximation and write the approximate equations of motion, and solve them for theta as a function of time.

Problem 3 Poisson Bracket, symmetry, conservation law

A system in 3D has Hamiltonian

H = $\frac{\vec{p}.\vec{p}}{2m} + \gamma\left(x^4 + 2x^2 y^2 + y^4\right)$ where $\vec{p} = \left(p_x, p_y, p_z\right)$, as usual.

Calculate the Poisson bracket of H with L_x, L_y, and L_z and interpret the result. What do these results imply about conservation of the various components of $\vec{L}$?

Problem 4 Normal modes

Three masses in a line (say, the x-axis) are connected by springs of equal spring constant k, but the assembly is otherwise free to move (not attached to walls). The two masses on the ends are of equal mass m, but the middle one is different, of mass M. Consider only motions along the x-axis.
a) write down the Lagrangian for this system in terms of coordinates x1, x2, x3
b) calculate the normal modes frequencies of vibration (eigenvalues) and modes (eigenvectors).
c) If you get a zero frequency mode, what does that mean?

Problem 5 Relativity

a) A 3-force $\vec{F}$ is applied to a relativistic particle moving at velocity/c of $\vec{\beta}$. The angle between $\vec{F}$ and $\vec{\beta}$ is 45 degrees. What is the angle between the resulting acceleration $\vec{a}$ and $\vec{F}$?
Express your result in terms of the mass *m, γ* etc.
b) Show, using the Euler-Lagrange equations, that the relativistic Lagrangian L= $\frac{-mc^2}{\gamma}$ gives the correct 3-vector equation of motion for a free particle, where, as usual, $\gamma = \left(1 - \vec{\beta}.\vec{\beta}\right)^{(-1/2)}$

Problem 6

a) A vat of fluid is uniformly rotated around a vertical axis at angular velocity ω in a uniform gravitational field g. Give an expression for the height of the fluid as a function of radial distance from the axis of rotation. Explain your equations and rationale clearly. Can you think of a use for this?
b) Using reasonable estimates of velocities and sizes, calculate the magnitude of the Coriolis force for water going down a sink drain. How does this compare with the gravitational force?